Hidden-State Probes, Answer Likelihood, and Surface Shortcuts in White-Box Hallucination Detection

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Abstract

Large language models can produce fluent but unsupported answers, motivating detectors that use internal model signals. We compare Base and Improved system profiles, then test added component groups on curated English TriviaQA hard negatives. Removing layer probes lowers mean deep-test AUROC by 0.0842 for Mistral-7B and 0.0958 for Llama-2-7B, while adding probes to the ablation Base arm raises it by 0.1479 and 0.1712. On English HaluEval, answer-only character TF–IDF reaches AUROC 0.9859, close to the hidden-state detector's 0.9931. On curated TriviaQA, the corresponding scores are 0.6076 and 0.8224. These contrasts locate the largest observed conditional gain in the probes and show that the shortcut risk depends on the benchmark.

Keywords: answer likelihood, hallucination detection, HaluEval, shortcut learning

1 Introduction

Large language models can produce fluent answers that contain unsupported claims. Detecting such answers matters when they are used to inform decisions, but a high benchmark score does not by itself show what a detector has learned. A detector can use the relationship between a question and its answer, internal model signals, or regularities in the answers' wording. The distinction matters for the reliability of hallucination detection in generative AI systems [1].

MultiHaluDet extracts features from a frozen language model, trains a deep detector, and combines predictions through a stacking ensemble [2]. The Improved configuration studied here adds answer-likelihood and verification features, layer probes, layer operations, a ranking loss, and a different stacking interface. A system-level comparison with Base measures their joint effect, but cannot attribute a difference to any one addition. We therefore examine what happens when each component group is added to a common control or removed from the complete detector. We also ask whether a detector's apparent success could be explained by answer-surface cues that do not establish factual support.

The first question asks how detection performance changes when each added component group is removed from the complete detector, with complementary add-one-group comparisons to clarify the control-to-full path. The second asks how well answer-surface models predict hallucination labels and how they compare with a detector that reads both the question and answer. We evaluate five seeds for Mistral-7B and Llama-2-7B on English HaluEval QA and manually curated TriviaQA hard negatives. Base denotes the comparison profile evaluated in this study, rather than the scores published for MultiHaluDet. The controlled component comparisons

use common question-grouped splits; the broader Base–Improved system comparison uses profile-specific splits and is interpreted accordingly.

To identify which additions matter, this study evaluates component groups on curated TriviaQA pairs through matched add-one-group and leave-one-group-out comparisons, with separate controls for the stacking interface. It then tests whether benchmark labels can be predicted from answer-surface cues by comparing shallow and answer-only text models with a question-and-answer detector on shared held-out rows within each dataset. The TriviaQA analysis uses a manually reviewed set of 4,952 hard-negative question pairs and also tests whether the detector ranks each hallucinated answer above its supported counterpart.

2 Related Work

Hallucination benchmarks differ in the errors they target and in how incorrect answers are collected [3]. TruthfulQA probes common misconceptions [4], whereas HaluEval labels question–answer records that include model-generated hallucinated answers [5]. If answer sources differ systematically in length, punctuation, or style, a classifier may predict the benchmark label without checking whether an answer follows from the question. Answer-only performance is therefore evidence of label predictability, not evidence that factual verification has succeeded.

Sampling-based detectors estimate uncertainty from generated responses. SelfCheckGPT measures consistency across samples [6], while semantic entropy groups responses by meaning before measuring uncertainty [7]. White-box methods instead inspect signals available inside a language model. Hidden-state probes can predict statement truthfulness, although sentence likelihood is also affected by length and word frequency [8]. INSIDE combines internal representations with response consistency [9]; InternalInspector draws on attention, feed-forward, and activation states [10]. LLM-Check combines hidden-state and uncertainty signals [1], and simple probes can approach multi-sample detectors at lower cost [11]. Prompt-guided internal-state methods such as PRISM address transfer beyond a single evaluation domain [12].

MultiHaluDet supplies the frozen-feature, deep-detector, and stacking pipeline examined in this work [2]. Its components make an aggregate improvement difficult to interpret without matched changes to the same pipeline. In particular, a gain from the complete Improved configuration cannot by itself identify whether likelihood features, probes, layer operations, or the stacking interface contributed. This motivates component-group comparisons that hold the data protocol fixed, alongside a system-level comparison that describes the combined configuration.

The shortcut question follows earlier hypothesis-only tests: natural-language-inference labels can be predicted without the premise because dataset construction leaves regularities in the hypothesis [13, 14]. Models can exploit such heuristics rather than the intended relation between inputs [15, 16]. An answer-only hallucination detector provides a similar diagnostic. Comparing it with a question-and-answer detector on the same held-out examples helps establish how much of the benchmark is predictable from the answer surface, while leaving the detector's remaining semantic contribution an open question.

3 Methodology

HaluEval uses the `qa_samples` configuration of `pminervini/HaluEval`; each record has a question, answer, and hallucination label. For TriviaQA, questions from the shuffled `lucadiliello/triviaqa` training split were answered by Gemma-2-2B-it with greedy decoding. Generated answers were retained only if they had at most 12 words and differed from every accepted alias after lowercasing and removing punctuation and articles, with substring matches also excluded. The first alias supplied the supported answer. We generated 5,000 initial question pairs, then manually checked their labels and answers. Manual review flagged 1,093 questions for issues involving a correct generated negative, an incorrect reference answer, reversed labels, a partially correct negative, text-normalization errors, stray markup or punctuation, or a time-sensitive answer. Depending on

the issue, the negative was regenerated, the label was manually adjudicated, or the text was corrected. Forty-eight context-dependent questions whose referents could not be resolved were excluded, leaving 4,952 pairs (9,904 records). The 1,093 flagged questions are not an additional set of exclusions; issue-specific counts and independent reviewer agreement are not reported. Lowercasing both answer types and stripping trailing periods reduced case and final-period cues in this dataset. The system-level comparison and the component-group ablation use this curated English set. The group ablation applies question-grouped splits to every condition.

Following dataset preparation, the framework extracts complementary hidden-state, probe-based, and answer-likelihood features from the frozen language models. These features are integrated into a dual-branch architecture, followed by out-of-fold stacking to produce the final hallucination predictions, as illustrated in Figure 1.

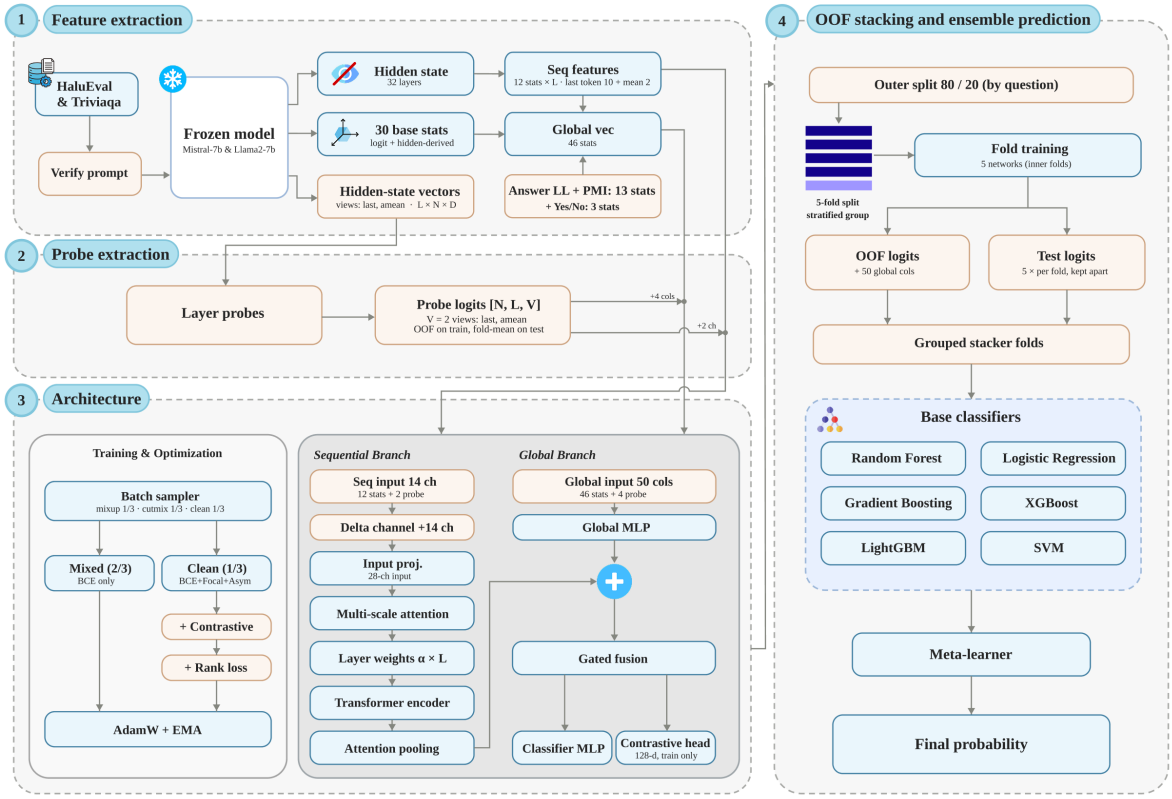


Figure 1: Orange blocks mark the added answer-likelihood features, layer operations, ranking loss, and logit stacking; green blocks mark the verification-prompt and layer-probe feature path. The remaining blocks show shared processing and evaluation stages.

3.1 Feature Extraction

Mistral-7B-Instruct-v0.2 and Llama-2-7B-hf are frozen feature producers. Base formats each record as `Question:` q and `Answer:` a, truncates to 256 tokens, and uses the hidden state after each of 32 transformer blocks. At every block, ten final-token statistics (norm, mean, standard deviation, minimum, maximum, positive and near-zero shares, softmax entropy, kurtosis, and median absolute deviation) and two mean-over-token statistics (norm and standard deviation) form a 32×12 sequence. Thirty global statistics summarize next-token outputs, changes in hidden-state norms across layers, anchor layers, and interactions.

Improved prompts for a Yes/No judgment using the question and proposed answer, truncates from the left at 384 tokens, and uses a question-free `Answer:` a pass as the likelihood reference for pointwise mutual

information (PMI). Answer-token offsets support mean pooling over answer tokens. Two hidden-state views are retained at every layer: the final prompt token and the mean answer-token state. For each view and layer, an outer-train probe applies standardization, PCA to 256 dimensions, and logistic regression ($C = 0.02$) in five question-grouped folds. Training rows receive out-of-fold logits; test rows receive the mean of the five fold probes. Two probe channels extend the layer sequence to 32×14, and four probe summaries extend the global vector. The resulting 50 global inputs comprise 30 base statistics, 13 answer-likelihood/PMI statistics, three Yes/No log-probability statistics, and four probe summaries.

3.2 Detector and Stacking

Table 1: Components of the system-level Base and Improved profiles. The Base arm of the component ablation uses question-grouped splits and the two-learner stacker, unlike the system-level Base profile.

Component	Base	Improved
Input prompt	`Question / Answer`	`Question / Proposed answer /` verification question (Yes or No)
Frozen-model passes	One	Verification pass + null pass (answer without question, for PMI)
Maximum length	256 tokens	384 tokens, truncated from the left
Mean-pooled statistics over	All input tokens	Answer tokens
Layer sequence	32×12 statistics	32×14 (12 statistics + 2 probe logits), 32×28 after layer differences
Global features	30	50 (30 base + 13 likelihood/PMI + 3 Yes/No + 4 probe summaries)
Layer probes	–	PCA-256 logistic probes on two views, grouped out-of-fold
Layer-difference channels	–	Yes
Layer-weight scaling	–	Yes
Pairwise ranking loss	–	Yes (weight 0.1)
Stacker input	Mean of the five 576-d representations	Fold logit + global features, applied per fold network
Stacker learners	Six	Two (logistic regression and LightGBM)
Data split and folds	Stratified by row	Grouped by question
Metrics	AUROC	AUROC + pair accuracy

Robust scaling of layer statistics and global features is fitted on the outer-training partition. The detector projects the layer sequence to 384 dimensions, combines windows of 1, 2, 4, and 8 layers through learned gates, and applies a six-layer Transformer with eight attention heads and dropout 0.25. Attention pooling yields a 384-dimensional layer representation; a two-layer global branch yields 192 dimensions. Their gated fusion produces a 576-dimensional vector, a classification logit, and a 128-dimensional contrastive projection. Improved also adds adjacent-layer differences (14 to 28 sequence channels) and rescales the learned layer weights to preserve their initial identity effect.

Five inner-fold detectors use AdamW (learning rate 4×10^{-4}, weight decay 6×10^{-5}), batch size 128, up to 45 epochs, five warm-up epochs, gradient clipping at 0.5, and an exponential moving average with decay 0.995. Mixup and CutMix each affect about one third of batches (Beta parameter 0.15). On unmixed batches, binary cross-entropy, focal loss, and asymmetric loss have weights 0.45, 0.35, and 0.20; supervised contrastive loss has weight 0.2 and temperature 0.04. Improved adds a pairwise AUROC surrogate with weight 0.1. Checkpoints maximize inner-validation AUROC, with patience 15. Selecting checkpoints on the fold later used for out-of-fold predictions can make those training outputs optimistic; the outer test is untouched.

Each fold network supplies its validation rows with out-of-fold vectors and logits and supplies test rows with

fold-specific outputs. The Base profile uses six stacker learners (random forest, histogram gradient boosting, LightGBM, logistic regression, XGBoost, and radial-basis support vector machine), whereas the Improved profile uses only logistic regression and LightGBM. Their five-fold out-of-fold scores feed a class-balanced logistic-regression meta-learner. Stacker folds use a distinct seed from detector folds. Base stacks 576-dimensional training vectors and averages the five test vectors before stacking. Improved stacks each detector logit with the global features and averages five test probabilities after stacking; its stacker folds are question-grouped. Table 1 summarizes the comparison.

3.3 Research Questions and Comparison Conditions

The component experiment on curated English TriviaQA includes an ablation Base arm, the full detector, three leave-one-group-out arms, and three add-one-group arms for each model at seeds 1, 11, 29, 37, and 43. Group A includes the verification prompt, 13 answer-likelihood/PMI features, and three Yes/No features; group B includes two layer-probe channels and four probe summaries; group C includes layer differences, layer-weight scaling, ranking loss, and the logit-stacking interface. All arms use question-grouped splits and the same two stacker learners, unlike the system-level Base profile in Table 1. Two additional interface controls use Base plus logit stacking (`add_C_logit`) and Full minus logit stacking (`loo_C_logit`); they do not add or remove the other parts of C. A separate probe-only logistic regression compares ordinary and verification prompts. Removing A also changes the prompt seen by the probes, so that contrast does not isolate answer likelihood.

The shortcut analysis compares shallow and answer-only models with the hidden-state detector on English HaluEval and curated TriviaQA. B0 uses answer length and its ratio to question length; B1 adds punctuation, casing, lexical statistics, question length, and question–answer overlap. Thus B0 and B1 are not strictly answer-only. B2 fits word- and character-level TF–IDF logistic regressions, while B3 fits XLM-R to answers alone. Vocabularies and transformations use outer-training rows only; within each dataset, model, and seed, the arms share outer-test rows. Complementarity and coarse length-bin analyses examine whether the hidden-state score contributes beyond these cues.

3.4 Controlled Experimental Design

The outer split separates training data from final evaluation rows, while inner folds support detector fitting and out-of-fold prediction generation. In the component ablation, each seed determines a question-grouped outer partition and training randomness; conditions at the same seed use the same split. The Improved configuration uses `GroupShuffleSplit`[1] for an 80/20 split by question and `StratifiedGroupKFold`[2] for the inner folds, layer probes, and stacker folds. Grouping prevents the two answers to one TriviaQA question from crossing a training, validation, or test boundary. The system-level Base profile instead uses a row-level split, so its comparison with Improved uses profile-specific outer-test rows. The component ablation holds the grouped data protocol fixed across its conditions, while variation across seeds combines data partitioning and training randomness.

For each condition and seed, let $y_i \in \{0,1\}$ be the outer-test label (1 for hallucinated) and s_i the predicted hallucination score. The primary metric, AUROC, is the probability that a randomly selected hallucinated answer receives a higher score than a supported answer, with half credit for ties. For N_+ hallucinated and N_- supported answers, it is

$$A = \frac{1}{N_+ N_-} \sum_{i:y_i=1} \sum_{j:y_j=0} \left[\mathbf{1}(s_i > s_j) + \tfrac{1}{2}\mathbf{1}(s_i = s_j) \right]. \tag{1}$$

For the grouped Improved TriviaQA runs, pair accuracy measures whether the hallucinated answer receives a higher score than the supported answer to the same question. Let $\mathcal{Q}$ contain outer-test questions retaining both

[1] This scikit-learn splitter assigns whole question groups to training or test data; the 80/20 proportion applies to groups, so the proportion of rows can differ.

[2] This scikit-learn splitter keeps question groups in separate folds while seeking similar class proportions across folds.

labeled answers, with scores $s_{q,1}$ and $s_{q,0}$; ties receive half credit:

$$P_{\text{pair}} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \left[\mathbf{1}(s_{q,1} > s_{q,0}) + \tfrac{1}{2}\mathbf{1}(s_{q,1} = s_{q,0}) \right]. \tag{2}$$

The row-split system-level Base profile does not preserve every question pair on one test partition, so no Base pair accuracy is compared in that analysis. Here $\mathbf{1}(\cdot)$ is the indicator function. For component group $g \in \{A, B, C\}$, a positive paired difference means that removing the group reduced AUROC; the mean and sample standard deviation over n completed seeds are

$$\Delta_g = \frac{1}{n} \sum_{r=1}^{n} (A_r^{\text{full}} - A_r^{\text{without g}}), \tag{3}$$

$$\bar{A} = \frac{1}{n} \sum_{r=1}^{n} A_r, \qquad s_A = \sqrt{\frac{\sum_{r=1}^{n} (A_r - \bar{A})^2}{n-1}}. \tag{4}$$

With five seeds, the reported differences are descriptive rather than conventional significance claims.

The shortcut analysis uses the same seed-summary formula in Equation (4). We report AUROC and, for grouped TriviaQA, pair accuracy; threshold-dependent metrics are omitted because thresholds were selected using outer-test labels. Complementarity and length-stratified results are descriptive. Diagnostic definitions appear in Appendix A, and the two feature-producing models are reported separately.

4 Results

Results first compare the Base and Improved system profiles on English HaluEval and curated TriviaQA, then isolates the contribution of each component group of Improved through ablation, and finally tests whether answer-surface shortcuts can account for the detector's performance on either benchmark. All results are reported as mean ± sample SD across five seeds.

Table 2: English HaluEval AUROC for the Base–Improved system comparison (mean ± sample SD across five seeds). Δ is the absolute difference between the multi-component Improved and Base profiles, which use profile-specific outer-test partitions.

LLM	Base AUROC	Improved AUROC	Δ
Mistral-7B	0.9849 ± 0.0030	0.9931 ± 0.0015	$+0.0082$
Llama-2-7B	0.9846 ± 0.0022	0.9917 ± 0.0013	$+0.0071$

The English HaluEval comparison shows higher mean AUROC for Improved, by 0.0082 for Mistral-7B and 0.0071 for Llama-2-7B. Its AUROC is higher in all ten model–seed comparisons. The profiles differ in prompt and probe features, split procedure, layer operations, ranking loss, and stacking interface. Their outer-test rows are not identical, so these descriptive differences belong to the combined profiles and cannot be assigned to one component.

Table 3: English TriviaQA hard-negative results on the curated 4,952-pair dataset (mean ± sample SD across five seeds). Δ compares the system profiles. Pair accuracy is available only for the grouped Improved runs.

LLM	Base AUROC	Improved AUROC	Δ	Improved pair acc.
Mistral-7B	0.6452 ± 0.0104	0.8226 ± 0.0040	$+0.1773$	0.8299 ± 0.0051
Llama-2-7B	0.5931 ± 0.0130	0.7650 ± 0.0062	$+0.1719$	0.7711 ± 0.0048

The curated English TriviaQA comparison shows mean AUROC differences of 0.1773 for Mistral-7B and 0.1719 for Llama-2-7B, and Improved is higher in all ten model–seed comparisons on profile-specific test partitions. The grouped Improved runs also rank the two answers to the same retained question, with mean pair accuracy 0.8299 and 0.7711, respectively. Base row-level splitting does not retain every question pair in one test partition, so no directly comparable Base pair accuracy is reported. The larger AUROC gaps on TriviaQA do not identify which of the simultaneous system changes caused them.

4.1 Component-Group Ablation of Improved

Table 4: Deep-test AUROC on curated English TriviaQA, computed from the mean of five fold-network logits (mean $\pm$ sample SD across five seeds). Addition arms start from the ablation Base arm; omission arms start from Full. These values are not stacked-ensemble AUROC.

Condition	Mistral-7B	Llama-2-7B
Base arm	0.6742 ± 0.0064	0.6043 ± 0.0089
Base + A	0.7392 ± 0.0122	0.6621 ± 0.0109
Base + B	0.8221 ± 0.0053	0.7756 ± 0.0039
Base + C	0.6706 ± 0.0081	0.6049 ± 0.0098
Full	0.8235 ± 0.0045	0.7653 ± 0.0062
Full without A	0.8221 ± 0.0057	0.7742 ± 0.0058
Full without B	0.7394 ± 0.0120	0.6694 ± 0.0132
Full without C	0.8213 ± 0.0040	0.7677 ± 0.0054

Using outer-test AUROC from the mean of five fold-network logits, allowing the component arms to be compared before stacking. This metric was chosen after observing large ensemble variation in vector-stacking arms; it does not measure the stacker-interface part of group C. Appendix Table 6 retains the corresponding ensemble results, including the two interface controls.

Adding B to the ablation Base arm raises mean deep-test AUROC by 0.1479 for Mistral-7B and 0.1712 for Llama-2-7B; removing B from Full lowers it by 0.0842 and 0.0958. Each paired contrast has the same direction in all five seeds. Adding A yields gains of 0.0650 and 0.0578, but removing A from Full changes AUROC by only $+0.0015$ and -0.0089 in full-minus-omission order. The latter contrast also changes the probe input prompt. Adding or removing C produces deep-test differences below 0.004; its stacking interface must be judged from ensemble results, not this metric. A probe-only logistic regression with the verification prompt reaches 0.8162 and 0.7590, versus Full deep-test AUROC of 0.8235 and 0.7653. With an ordinary prompt, probe-only AUROC is 0.8262 and 0.7811; because the prompt differs, these are separate comparators rather than a controlled ablation of the complete detector.

4.2 Answer-Surface Predictability Across Benchmarks

The hidden-state detector reaches a much higher AUROC on HaluEval than on curated TriviaQA. This gap could arise if HaluEval labels can be predicted from the answer text alone, without checking whether the answer is supported. Accordingly, two surface cues are examined in English HaluEval: answer length and answer end with a period.

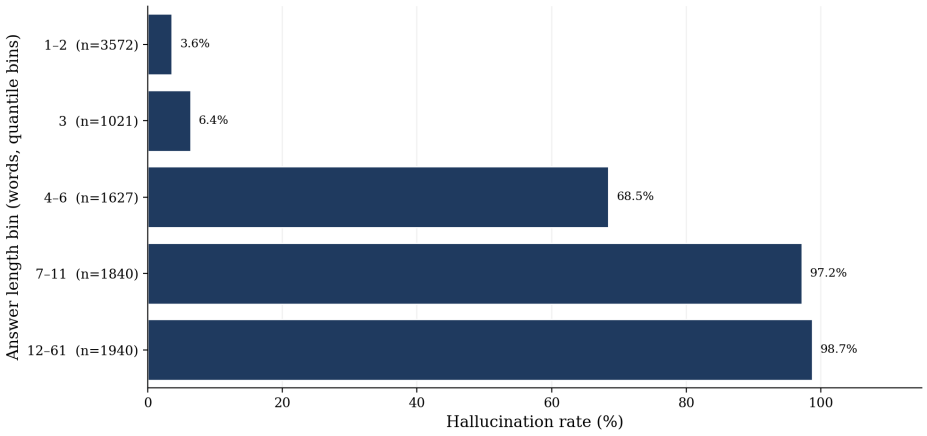


Figure 2: Hallucination rate across answer word-count groups in English HaluEval. Bin labels include the number of answers in each group.

This figure shows hallucination rates of 3.6% for one- or two-word answers and 6.4% for three-word answers. The rate rises to 68.5% for four- to six-word answers and to 97.2% and 98.7% in the two longest groups. Supported answers are concentrated in the shorter groups, whereas hallucinated answers are concentrated in the longer groups. Consistent with this pattern, answer character count alone yields AUROC values of 0.9721 for Mistral-7B and 0.9706 for Llama-2-7B.

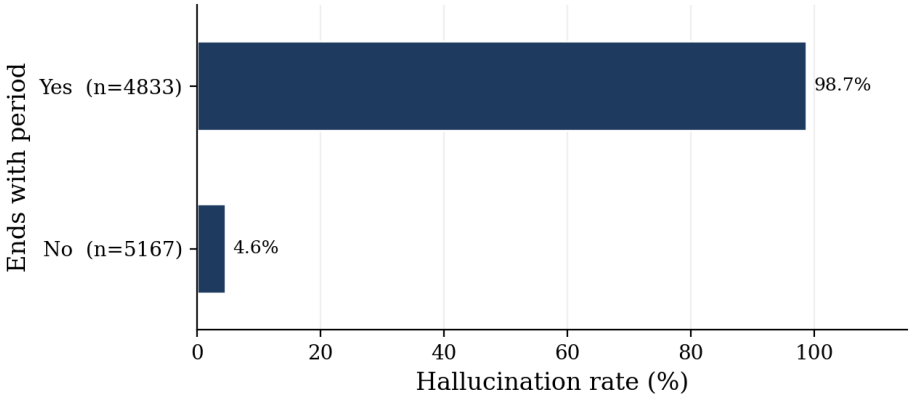


Figure 3: Hallucination rate in English HaluEval by whether an answer ends with a period. Category labels include the number of answers.

Among 4,833 answers with a terminal period, 98.7% are labeled hallucinated, compared with 4.6% of the 5,167 answers without one. The terminal-period feature alone yields AUROC values of 0.9726 for Mistral-7B and 0.9704 for Llama-2-7B; permuting it in B1 reduces AUROC by 0.0974 and 0.0800, respectively. Neither answer length nor a terminal period establishes factual support. These patterns are consistent with differences in how the two answer types were constructed.

Table 5: AUROC on English HaluEval and curated TriviaQA (mean ± sample SD over five seeds). B0 and B1 include question-derived covariates; B2 and B3 use answer text alone. The XLM-R score describes this particular training setup, not an upper bound on answer-only models.

Dataset	Detector	Mistral-7B	Llama-2-7B
HaluEval	B0 shallow (GBM)	0.9755 ± 0.0027	0.9747 ± 0.0044
	B1 shallow (GBM)	0.9888 ± 0.0019	0.9872 ± 0.0028
	B2 answer char	0.9859 ± 0.0010	0.9851 ± 0.0015
	B3 XLM-R answer	0.9870 ± 0.0008	0.9860 ± 0.0006
	Hidden-state final	0.9931 ± 0.0012	0.9919 ± 0.0014
TriviaQA	B0 shallow (GBM)	0.5690 ± 0.0163	0.5699 ± 0.0173
	B1 shallow (GBM)	0.5794 ± 0.0115	0.5807 ± 0.0125
	B2 answer char	0.6076 ± 0.0102	0.6076 ± 0.0102
	B3 XLM-R answer	0.5127 ± 0.0230	0.5127 ± 0.0230
	Hidden-state final	0.8224 ± 0.0038	0.7654 ± 0.0063

Table 5 contrasts representative surface models with the final hidden-state detector on identical outer-test rows within each dataset, model, and seed. On HaluEval, answer-only character TF–IDF and XLM-R nearly reach the detector, while a question-only TF–IDF control remains near chance. On curated TriviaQA, answer-only character TF–IDF reaches 0.6076 AUROC for both feature-producing models, substantially below the detector's 0.8224 and 0.7654; question-only scores are 0.5 by the paired construction. For TriviaQA, the identical B2 and B3 values across the two language-model columns come from the same answer-only text evaluation; the HaluEval values differ and are reported separately.

On HaluEval, answer-only surface detectors nearly match the hidden-state detector. Length alone (B0) reaches AUROC 0.9755 and 0.9747 for Mistral-7B and Llama-2-7B, B2 and B3 reach 0.985-0.987, and B1 reaches 0.9888 and 0.9872, within 0.0043 and 0.0047 of the hidden-state detector (0.9931 and 0.9919). The best surface baseline reproduces 99.1% and 99.0% of the detector's above-chance signal, so a score above 0.99 is compatible with answer-construction artifacts and does not by itself indicate that factuality is read from the hidden states.

On curated TriviaQA, surface baselines stay close to chance (B0 0.569-0.570, B1 0.579-0.581, B2 0.608, B3 0.513), whereas the hidden-state detector reaches 0.8224 on Mistral-7B and 0.7654 on Llama-2-7B. The best surface baseline reproduces only 33.4% and 40.5% of this signal. Surface leakage is therefore about 99% on HaluEval and about 33-40% on TriviaQA; these descriptive comparisons do not identify a semantic mechanism or establish cross-seed significance.

5 Discussion

The Base–Improved system comparison favors Improved on both datasets, but it changes the prompt, features, split, training, and stacker together. The component experiment provides narrower evidence: adding probe group B to the ablation Base arm produces the largest deep-test AUROC gain, and removing B from Full produces the largest decline. The A contrast is less clean because removing A changes the prompt used by the probes. Deep-test AUROC cannot evaluate the stacking interface in C; ensemble results and interface controls are reported separately. Sentence likelihood itself depends on length and word frequency [8], so a gain from answer-conditional inputs would not alone demonstrate factual verification.

The answer-only baselines nearly reach the hidden-state detector on HaluEval but remain far below it on curated TriviaQA. The discrepancy makes shortcut risk specific to benchmark construction rather than a property established for every hallucination dataset. Coarse length bins leave substantial length predictability on HaluEval, and the near-chance question-only control cannot rule out question–answer interactions. Similar artifact-driven behavior has been documented in natural-language inference [14, 15]; reporting stratified scores helps expose

such patterns [17]. The remaining TriviaQA advantage warrants further study but does not establish that the detector verifies meaning.

5.1 Limitations and Threats to Validity

The five seeds limit precision and do not support a conventional two-sided Wilcoxon claim: its smallest attainable p-value is 0.0625. A seed changes both the outer split and training randomness, so their variances cannot be separated. The negative answers come from one generation procedure. Checkpoints are selected on the fold later used for detector out-of-fold outputs, which can make those outputs optimistic. Probe and detector folds also differ, so probe fitting can use labels from a detector-validation fold even though each probe excludes its own question pair. These procedures are shared within relevant paired arms, but bias cancellation has not been tested; the apparent absolute probe contribution may be overstated. Outer-test labels are not used to fit the detector. The system-level comparisons also use profile-specific test partitions, and HaluEval AUROC is close to the ceiling.

Answer-only scores can reflect generation-source style, length, punctuation, or templates rather than factual support. B0 and B1 include question-derived covariates. Five broad length bins leave residual length information on HaluEval, where B0 still reaches matched AUROC 0.8393; no equivalent exact-length control establishes semantic independence on TriviaQA. The XLM-R baseline is not an upper bound on answer-only prediction. Four Llama-2-7B feature rows were absent from the HaluEval shortcut runs, leaving 7,996 training rows and 2,000 test rows. These analyses do not isolate a semantic hidden-state signal from all surface cues.

6 Conclusion

The curated TriviaQA ablation identifies probe group B as the largest observed conditional contributor: adding it to the ablation Base arm raises mean deep-test AUROC by 0.1479 and 0.1712, and removing it from Full lowers AUROC by 0.0842 and 0.0958 for Mistral-7B and Llama-2-7B. The answer-conditional group does not show a consistent removal effect across models. These results concern the evaluated grouped protocol; checkpoint reuse, differing folds, and the prompt change in the A contrast limit causal interpretation.

Answer-only text nearly predicts HaluEval labels as well as the hidden-state detector, while on curated TriviaQA answer-only character TF–IDF reaches AUROC 0.6076 versus detector scores of 0.8224 and 0.7654. Length and terminal punctuation remain strong HaluEval cues. The cross-benchmark contrast reveals a substantial shortcut risk in HaluEval without proving that the remaining TriviaQA advantage represents semantic verification.

Future work should evaluate the detector on benchmarks whose construction weakens answer-surface cues. A length-controlled version of HaluEval, in which supported and hallucinated answers are matched in length and format when the dataset is built, would show how much detection performance remains once the dominant shortcuts are removed. Evaluation on TruthfulQA [4], on the naturally generated and word-level annotated responses of RAGTruth [18], and on the dialogue and summarization tasks of HaluEval would test external validity beyond synthetic question answering. Comparisons with single-pass hidden-state probes [11] and with sampling-based detectors such as semantic entropy [7] would place the pipeline among established methods. Finally, the effect of NF4 quantization and batched extraction on feature fidelity, which was not examined here, could be measured by comparing full-precision and quantized extraction on identical examples.

A Reproducibility Details

For the shortcut analysis, we retained per-seed TriviaQA test scores, labels, and row indices, allowing AUROC and pair accuracy to be recomputed from predictions. For the component ablation, we retained run-level aggregates but not row-level ensemble probabilities, so those metrics cannot be independently recomputed from predictions. Resolved configurations, source revisions, training-row identities, and logs are also incomplete for a

full rerun. Independent reconstruction would require the fold-specific logits and final ensemble probabilities together with labels and row identities.

Table 6: Stacked-ensemble AUROC on curated English TriviaQA (mean ± sample SD over five seeds). The two logit-interface controls alter only the stacker input relative to Base or Full; other addition and omission arms alter whole component groups.

Condition	Mistral-7B	Llama-2-7B
Base	0.5230 ± 0.1682	0.4937 ± 0.0995
Base + A	0.6418 ± 0.2009	0.5908 ± 0.1247
Base + B	0.6867 ± 0.2861	0.4518 ± 0.2909
Base + C	0.6710 ± 0.0090	0.6062 ± 0.0102
Base + logit interface	0.6737 ± 0.0079	0.6043 ± 0.0098
Full	0.8226 ± 0.0040	0.7650 ± 0.0062
Full without A	0.8199 ± 0.0055	0.7729 ± 0.0057
Full without B	0.7391 ± 0.0110	0.6717 ± 0.0114
Full without C	0.2787 ± 0.1783	0.5450 ± 0.2848
Full without logit interface	0.5581 ± 0.3444	0.3202 ± 0.1038

Table 6 reports stacked-ensemble AUROC for the eight component arms and two interface controls. Base, Base + A, Base + B, Full without C, and Full without logit interface use vector stacking. Base + C and Full use logit stacking. The `add_C_logit` and `loo_C_logit` controls isolate the interface relative to Base and Full, respectively. Large dispersion in vector arms is compatible with fold-representation misalignment, but these results do not isolate that mechanism.

In the diagnostic analyses, Spearman correlation is Pearson correlation of the average ranks of feature x and score z. Permutation decrease compares original AUROC A_0 with the mean after $K = 10$ test-row shuffles of feature j. For weighted AUROC, valid length bins $\mathscr{B}$ have at least ten rows per label, with bin size n_b and within-bin AUROC A_b. For matched AUROC, each of $R = 20$ repetitions balances labels within every bin containing both classes before pooling.

$$\rho_S = \mathrm{corr}(\mathrm{rank}(x), \mathrm{rank}(z)), \tag{5}$$

$$I_j = A_0 - \frac{1}{K} \sum_{k=1}^{K} A_{\pi_j^{(k)}}, \tag{6}$$

$$A_{\text{weighted}} = \frac{\sum_{b \in \mathscr{B}} n_b A_b}{\sum_{b \in \mathscr{B}} n_b}, \tag{7}$$

$$A_{\text{matched}} = \frac{1}{R} \sum_{r=1}^{R} A(y^{(r)}, s^{(r)}). \tag{8}$$

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